

		'stationary' from the disc's perspective)
25	D	Since transformer has turn ratio of 2, secondary voltage is 240V. Hence power consumed is $P = \frac{(240)^2}{8} = 7200W$ Since transformer has efficiency of 0.75, power drawn from a.c. supply is $P_{ac} = \frac{7200}{0.75} = 9600W$ Hence current in primary coil is given by $I = \frac{P_{ac}}{V} = \frac{9600}{120} = 80A$
26	B	Since $P \sim V^2$ if kettle is connected to mains voltage that is half the recommended value, its